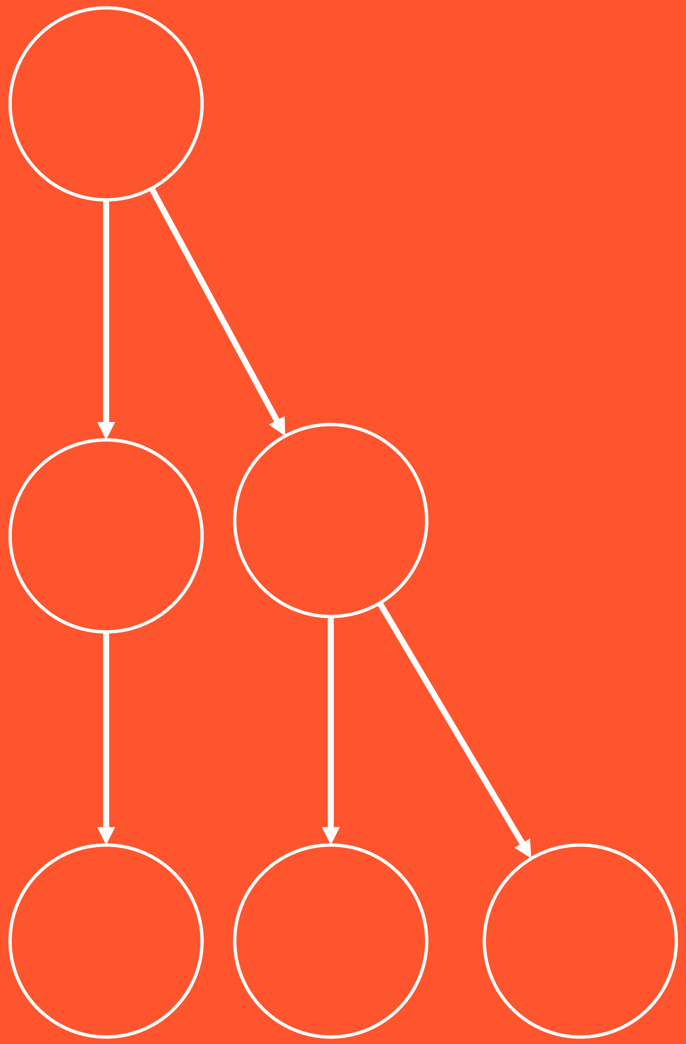


BCOG 100

Course Overview

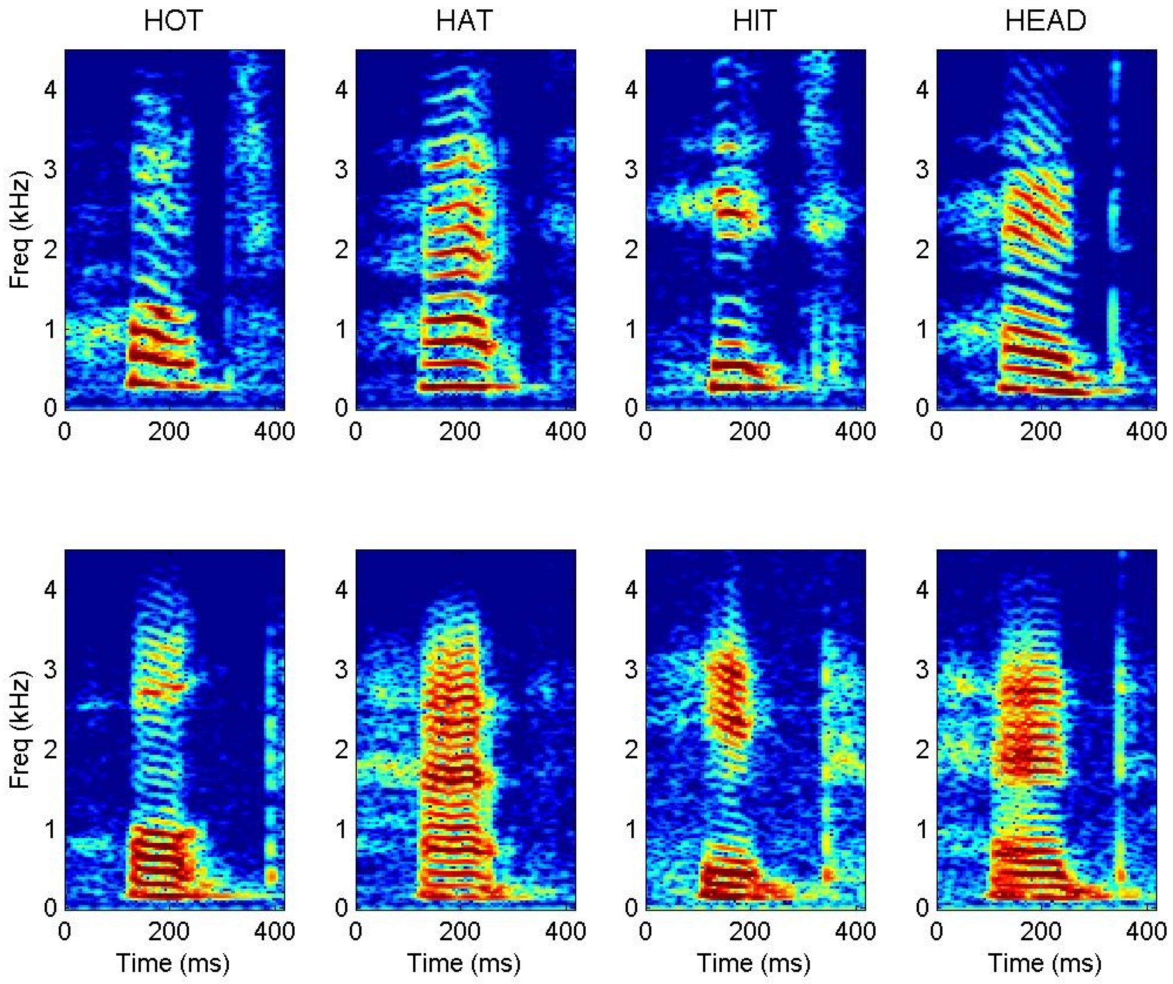
Lecture 0 — What Is Brain and Cognitive Science?

UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

Look At What You Can Do

Look At What You Can Do

Look At What You Can Do



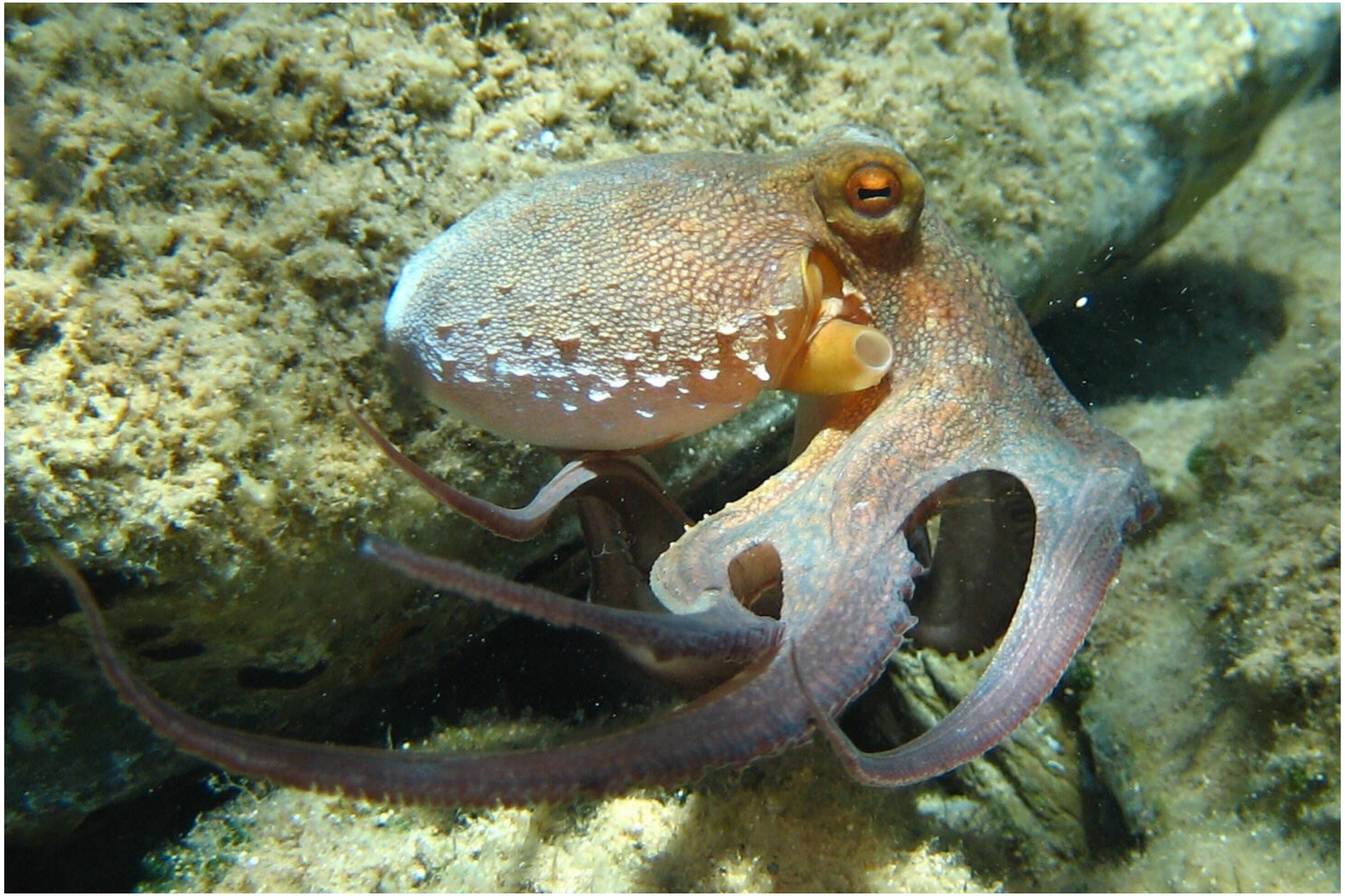
Look At What You Can Do

And This One Is Three Years Old

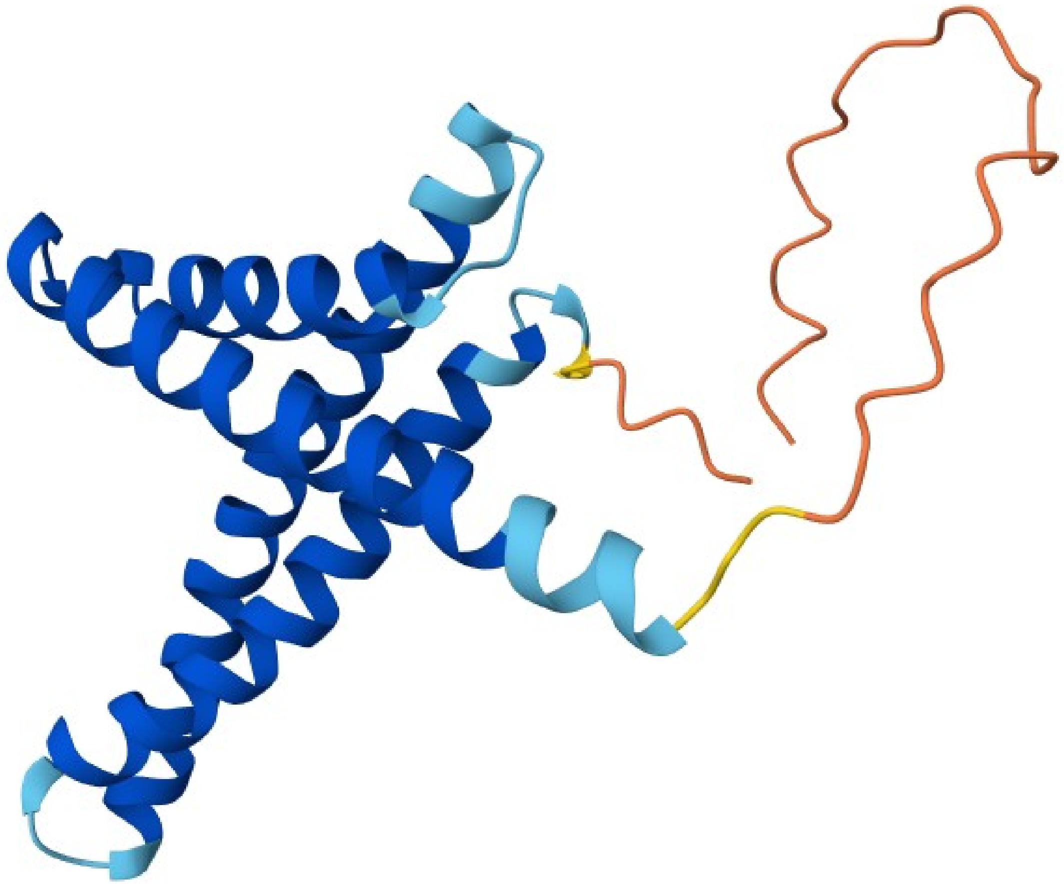
Minds Built Nothing Like Ours

Minds Built Nothing Like Ours

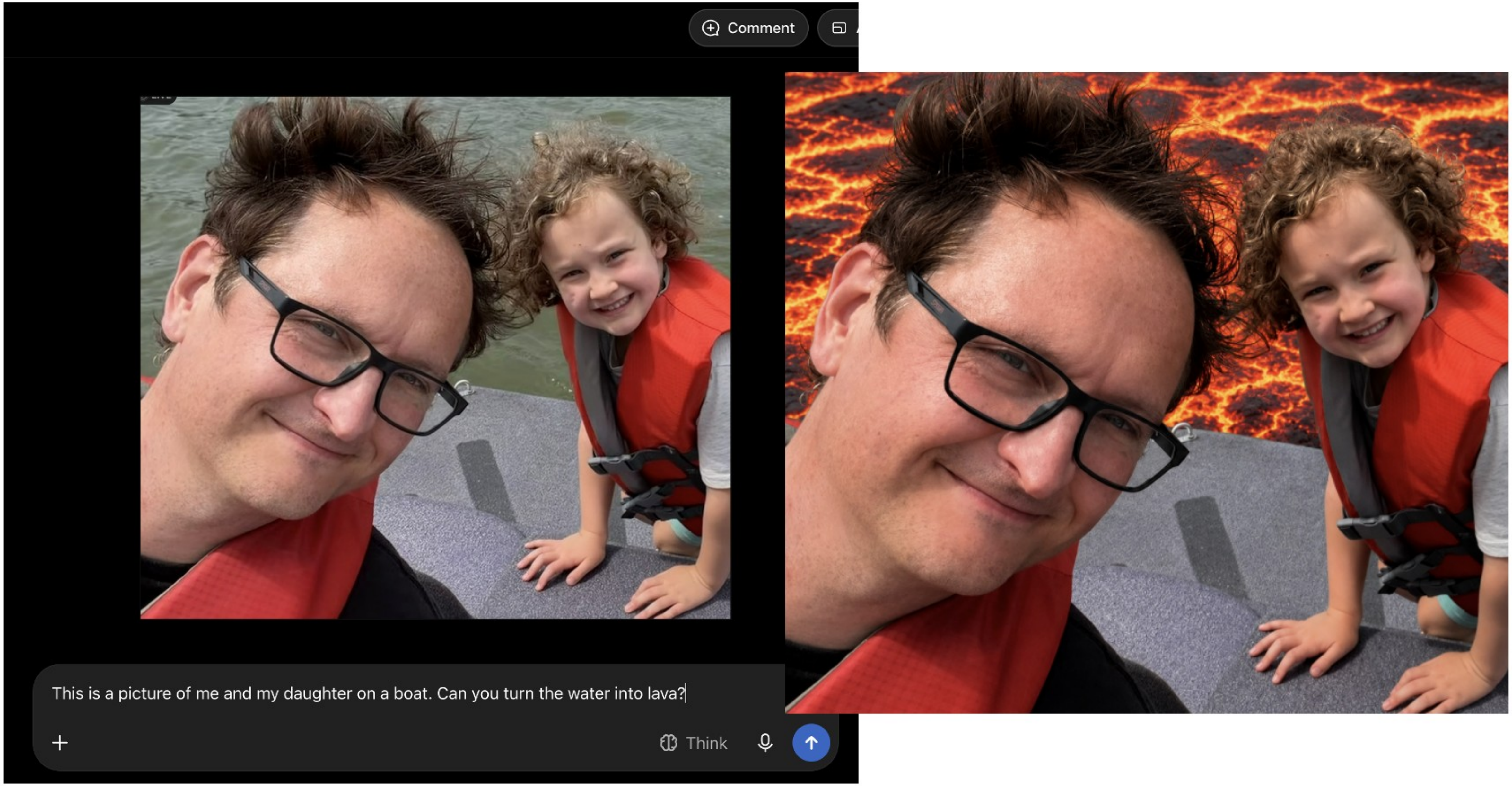
Minds Built Nothing Like Ours



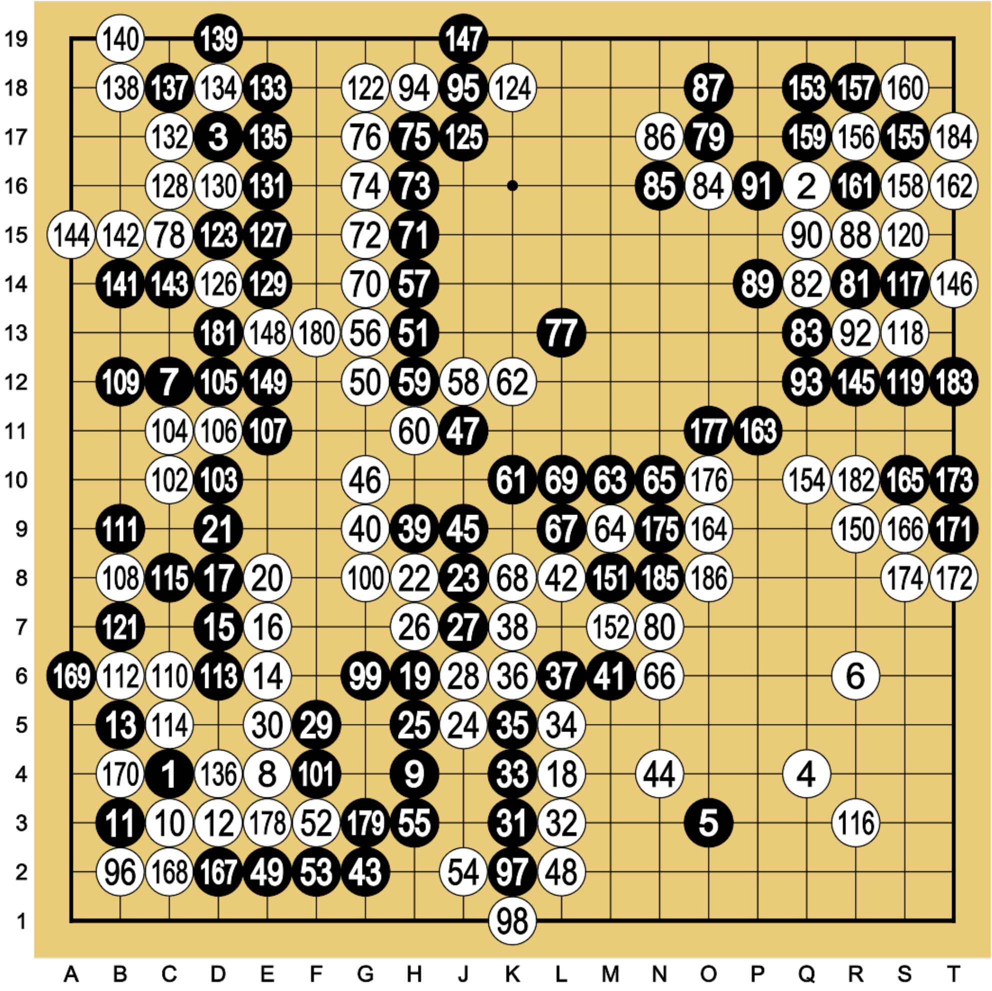
Minds With No Neurons At All



Minds With No Neurons At All

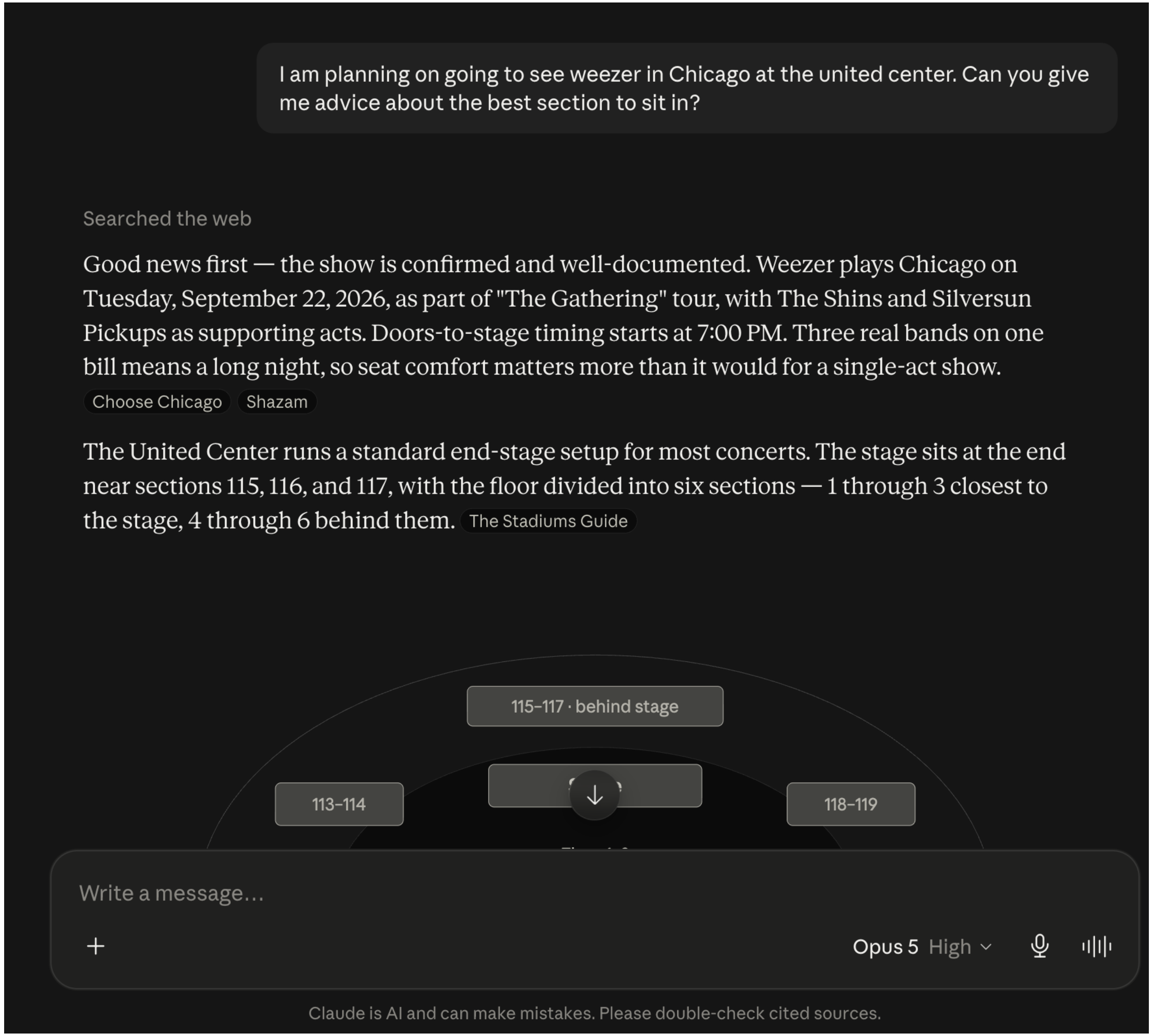


Minds With No Neurons At All



Lee Sedol (B) vs AlphaGo (W) - Game 1

Minds With No Neurons At All



Twenty Watts

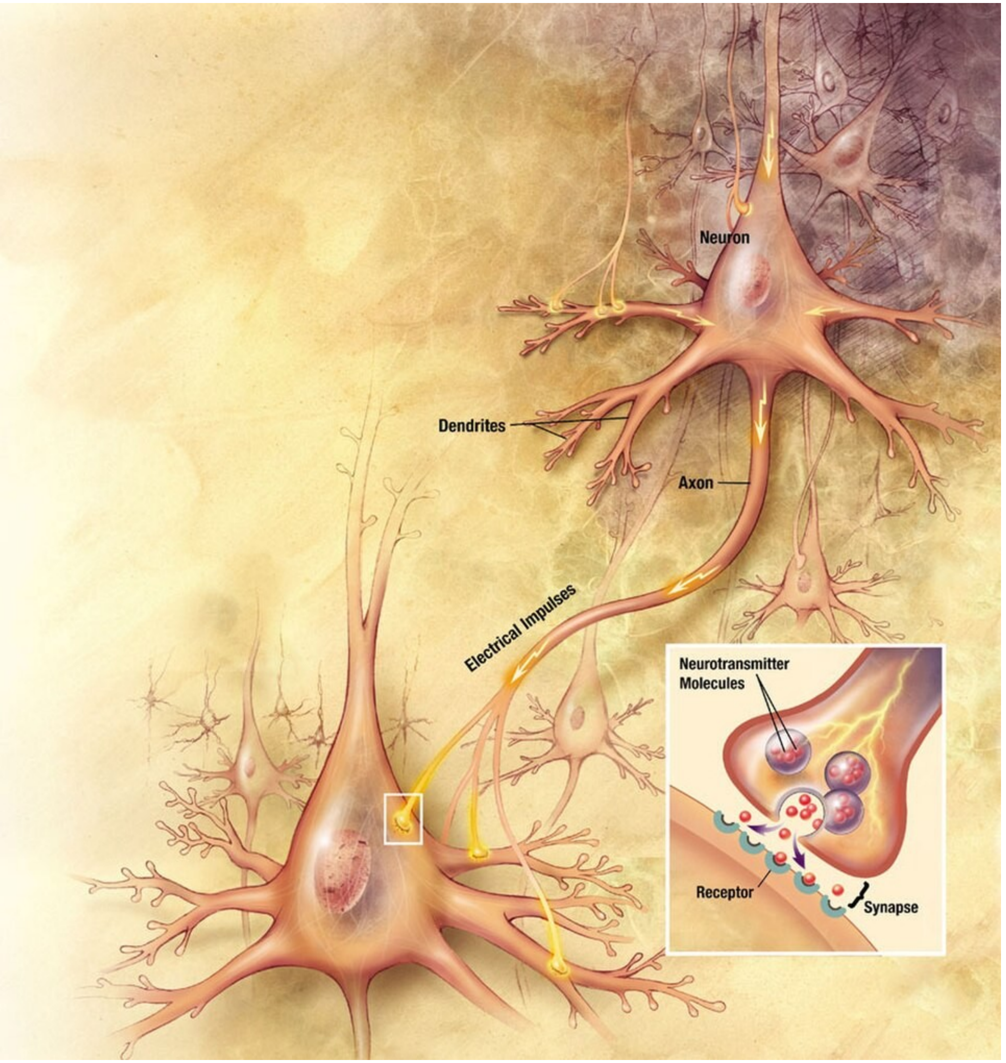
One neuron is a fairly simple machine

You have 86 billion of them, each in conversation with thousands of others

The whole thing runs on about 20 watts — less than a household lightbulb

The machines that fold proteins, play Go, and run chatbots fill warehouses and draw the power of a small town

How do you get all of that out of this?



Twenty Watts

One neuron is a fairly simple machine

You have 86 billion of them, each in conversation with thousands of others

The whole thing runs on about 20 watts — less than a household lightbulb

The machines that fold proteins, play Go, and run chatbots fill warehouses and draw the power of a small town

How do you get all of that out of this?

Which of These Is Most Impressive?

Which of the things we just saw is the most impressive?

Which one is least like the others?

Which of them, if any, has a mind?

What Is Brain and Cognitive Science?

The study of the mind and the brain, with two commitments:

1. Interdisciplinary — no one field can explain a mind by itself
2. Computational — minds take in information, store, represent, or are affected by the info, and act on it

An ambition more than an accomplished fact

Interdisciplinary BCS

Psychology

Neuroscience

Philosophy

Computer Science and Artificial Intelligence

Anthropology

Linguistics

Education

Ecology and Evolutionary Biology

One Moment, Five Questions

A three-year-old says: "I goed to the store."

Neuroscientist — what has to be in place, physically, for her to produce a form she has never once heard?

Psychologist — did she look up a word, or did she build something and apply it?

AI researcher — what would a system need, in order to make exactly this mistake?

Linguist — "goed" is nowhere in what she heard. What does the error reveal about the grammar she has already built?

Philosopher — is "she has a rule" a claim about what is in her head, or just a tidy description of her behaviour?

Computational BCS

Computational Metaphor

- Information comes in, is held or transformed by the system somehow, and an action comes out
- A common shorthand: the brain is the hardware, the mind is the software
- Two questions follow, and they run through the whole course:
  - how does a mind hold on to information, and how does it act on what it holds

Computational Methods

- Computer-based experiments
- Computer-driven tools like EEG and fMRI
- Complex data analysis and machine learning
- Computational models and simulations

Is the Brain Really a Computer?

The brain has no clean divide between hardware and software

It does not run in the tidy, step-by-step way a laptop does

A mind is bound up with a body and a world, not sealed inside a skull

"Computation" may have stretched so far that it explains nothing

A lens, not a doctrine

Three Levels, Three Questions

What problem is this system solving, and why?

How does it solve that problem — what does it represent, and what processes act on it?

What is it physically made of, and how does that stuff carry the work out?

Not three rival answers. Three different questions.

Where We're Going

We follow the order evolution built things in — and at each step, we compare against machines.

What evolution added	The problem it solved
The first neurons	Sense and act faster than chemistry alone allows
Neural circuits, affect and valence	Work out whether a thing is good or bad, and do something about it
Learning and plasticity	Change with experience, instead of being finished at birth
The vertebrate brain	Run many systems at once, in one body
Pattern recognition	Find the same thing again, having never seen it the same way twice
Perception and action	Act on a world you can only ever sample
Spatial cognition	Know where you are, and get back
Memory, imagination and generative models	Use the past, and run a future before living it
Social cognition	Predict the most complicated thing in the environment: other minds
Planning and decision-making	Choose between futures that have not happened
Language	Put a thought into someone else's head
Symbolic cognition	Think about what is not present, and what is not real
Culture	Inherit what others worked out, without working it out yourself

What a Week Looks Like

The reading is posted before the week it covers begins

Three lectures

One lab

A five-minute quiz at the start of most classes — starting in week 2, and always about the week before

Practice questions, ungraded and unlimited, on Canvas

How You're Graded

Daily in-class quizzes — 136 points, 34% (40 given, 34 counted)
Lab assignments — 120 points, 30% (14 given, 12 counted)
<u>Final exam — 144 points, 36%</u>
Total — 400 points

Extra credit: research participation (SONA) up to 3%, course notes up to 3%

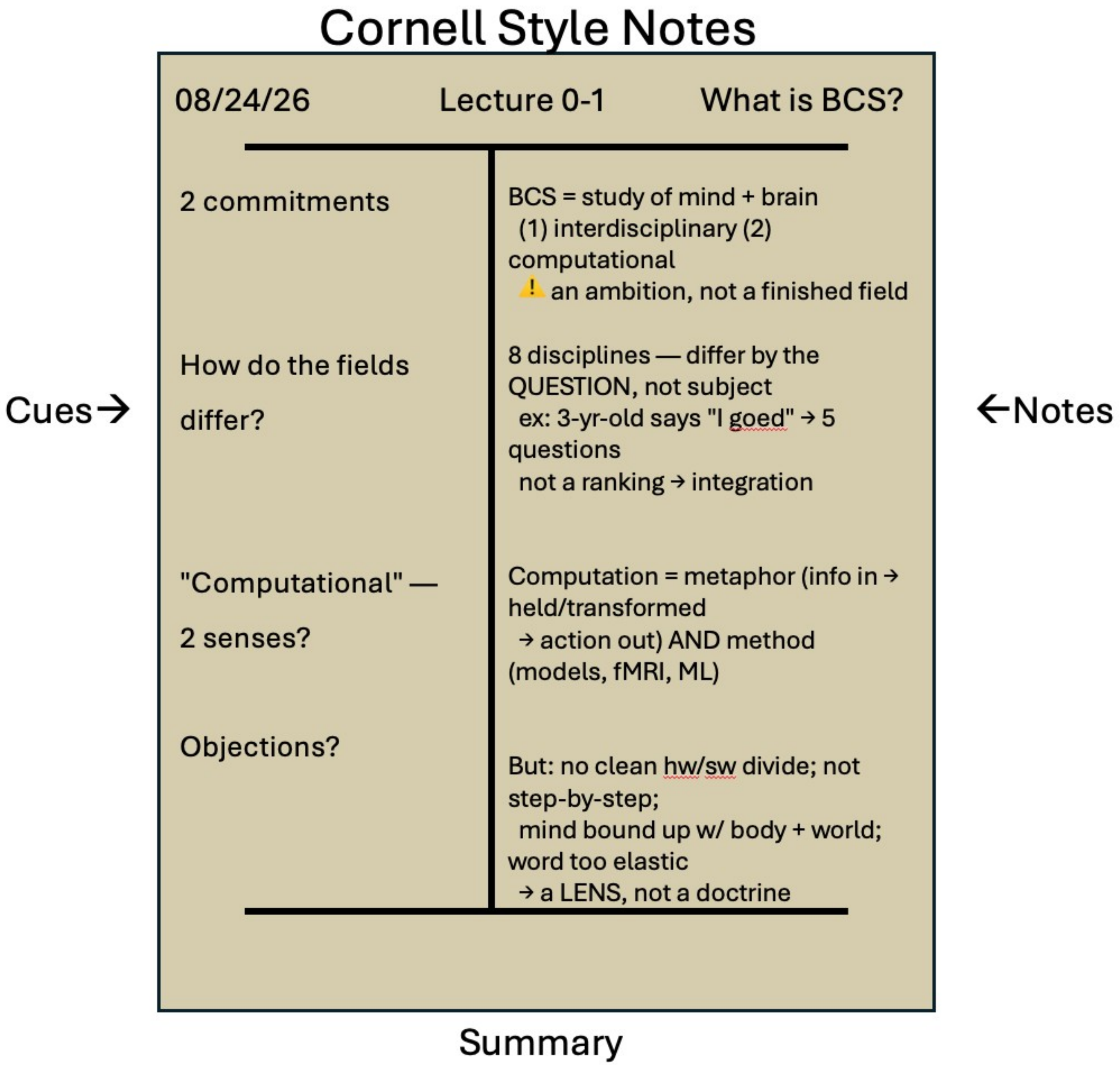
Generous drops, which is why there are no make-ups

Why It's Built This Way

- Practice questions on Canvas: ungraded, unlimited, with explanations for every answer
- The daily quiz is on paper, closed book, drawn from those same practice questions
- Quizzes start in week 2, and each one covers the week before — you always get a week with the material first
- The final is drawn from them too
- No midterm — the daily quizzes already spread the observations out

How to Do Well

Come to class. Do the reading before the week starts. Work the practice questions.
Take notes with a real method — SQ3R, Cornell Notes — described in a document on Canvas.
Ask questions, and come to office hours



How to Do Well

Beware the illusion of understanding — following an explanation is not the same as being able to produce it

The fix is not rereading. It is closing the thing and making yourself produce the answer, badly, before you check

Using AI in This Course

Not this. Do not have a model produce answers, text, code, or figures you hand in. That covers every lab

This, and I encourage it. Re-explain a passage you have already read · quiz you, answer withheld until you commit · decode notation · argue with reasoning you have already worked out

The test: if you finish and could not now do the thing yourself, it did not help you

But it doesn't have to go that way:

Used intentionally, ChatGPT can actually **make you smarter** by:

- Acting as a **sounding board** to clarify your thoughts.
- Offering **different perspectives** or ideas you hadn't considered.
- Helping you **learn faster** by breaking down complex ideas.
- Letting you test and refine your own reasoning by comparing it to an alternative.

It's kind of like Google, Wikipedia, or even books: they **can be tools for learning**, or they can be shortcuts that undermine it—it depends **how** you use them.

So, what's the difference between smart use and dumb use?

Smart Use	Dumb Use
Asking for explanations, then reflecting or researching further	Blindly copying answers
Using ChatGPT to test your understanding or debate ideas	Letting it do all the work
Treating it like a collaborator or tutor	Treating it like a crutch

Using AI in This Course

These tools **raise the ceiling** for someone who already has the skill and **lower the floor** for someone who does not — for the same reason. An expert is checking the output; a novice is trusting it

Experienced developers, on their own job: expected **24% faster** · believed they *had been* **20% faster** · measured **19% slower**

Full policy in the syllabus · the whole argument in *Artificial Intelligence in This Course*, on Canvas · each lab report ends with one sentence on whether you used AI

Take Care of Yourself

Sleep, eat, move, keep a schedule you can actually keep

Wellbeing, writing, and disability resources are linked on the syllabus

Asking early works far better than asking late

Before Next Class

Read Chapter 1 — Mind and Brain

The question we open with: is your mind nothing but your brain?

Image Sources (1 of 2)

Slide	Image	File	Licence	Source
2	Green Street, Champaign	green_street_crowd.jpg	CC BY 4.0	Jon Willits — own photograph · own work
2	Sliced oranges	oranges.jpg	CC BY-SA 4.0	Edithobayaa1 · https://commons.wikimedia.org/wiki/File:Sliced_oranges_in_Ghana.jpg
2	Spectrograms: HOT, HAT, HIT, HEAD	spectrograms_hot_hat_hit_head.jpg	CC BY 4.0	Jon Willits — own figure · own work
2	Distracted boyfriend meme	distracted_boyfriend_meme.jpg	All rights reserved — used as fair use, educational; licence being sought	Photograph by Antonio Guillem (to verify) · https://commons.wikimedia.org/wiki/File:Distracted_boyfriend_meme.jpg
3	Children building together	children_block_tower.png	CC BY 4.0	Jon Willits — generated with ChatGPT (OpenAI) · own work
4	Arctic tern in flight	arctic_tern_in_flight.jpg	CC BY-SA 4.0	Charles J. Sharp, Sharp Photography · https://commons.wikimedia.org/wiki/File:Arctic_tern_(Sterna_paradisaea)_in_flight_Myrar.jpg
4	The honeybee waggle dance	honeybee_waggle_dance.png	CC BY 2.5	Figure design: J. Tautz and M. Kleinhenz, Beegroup Würzburg; from Chittka L, PLoS Biol 2004 · https://commons.wikimedia.org/wiki/File:Bee_waggle_dance.png
4	Common octopus	octopus.jpg	CC BY-SA 3.0	Beckmannjan · https://commons.wikimedia.org/wiki/File:Octopus_vulgaris2.jpg

Image Sources (2 of 2)

Slide	Image	File	Licence	Source
5	AlphaFold prediction — TIMM17A	alphafold_predicted_structure.png	CC BY 4.0	AlphaFold Protein Structure Database (Jumper et al., 2021) · https://commons.wikimedia.org/wiki/File:Prote%C3%ADna_TIMM17A.png
5	Turning water into lava	ai_image_editing.png	CC BY 4.0	Jon Willits — own photograph, edited with ChatGPT (OpenAI) · own work
5	Lee Sedol vs AlphaGo, game 1	alphago_board.png	CC BY-SA 4.0	Wesalius · https://commons.wikimedia.org/wiki/File:Lee_Sedol_(B)_vs_AlphaGo_(W)_-_Game_1.svg
5	A conversation with a chatbot	chatbot_interface.png	CC BY 4.0	Jon Willits — own screenshot; interface is Claude (Anthropic) · own work
6	Neurons and the synapse	neurons_wikimedia_commons.jpg	Public domain (PD-USGov)	https://commons.wikimedia.org/wiki/File:Neurons_big1.jpg
6	CERN data centre	datacenter_hall.jpg	CC BY-SA 3.0	Florian Hirzinger · https://commons.wikimedia.org/wiki/File:CERN_Server_03.jpg
18	Cornell Notes — this lecture	cornell_notes_example.png	CC BY 4.0	Jon Willits — own work · own work
19	ChatGPT on smart use versus dumb use	llm_screenshot_2.png	Author's own screenshot; the interface shown is the vendor's	Screenshot by Jon Willits · own work